

Emergent Quantum Gravity on D_4 : Visons, the Continuum Einstein-Hilbert Limit, and One-Loop Finiteness

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May 2026

Abstract

We develop a discrete quantum-gravitational theory on the D_4 root lattice, with three-dimensional FCC layers as constant-time spatial slices. Within the [[192, 130, 3]] CSS stabilizer code of [4], we identify the propagating graviton as a coherent plane-wave superposition of *visons*: localized m-type code excitations obtained by ceasing to measure one octahedral-void X -stabilizer. We motivate this architectural choice axiomatically: among all local discrete excitations of the FCC stabilizer code, the vison is the unique candidate satisfying the five defining properties of a graviton (massless dispersion, spin 2, gauge-invariant content, universal coupling, and continuum reduction to linearized Einstein-Hilbert). A single removed octahedral-void stabilizer produces Regge deficit $\arccos(-1/3) \approx 109.47^\circ$ at each of 12 surrounding edges; the Roček-Williams theorem maps the linearized Regge action on D_4 to the linearized Einstein-Hilbert action, giving $\omega = c|\vec{k}|$, masslessness, and exactly two transverse-traceless spin-2 polarizations. Newton's constant $G = a^2/(8 \ln 2)$ is fixed by Bekenstein-Hawking entropy matching to the 2D sheet-plaquette area $L_0^2 = a^2/2$. The Cheeger-Müller-Schrader theorem applied to D_4 gives the nonlinear continuum limit: discrete Regge $\rightarrow$ Einstein-Hilbert, with the rank-four bond-tensor isotropy $T^{(4)} = 4S^{(4)}$ ensuring that all $O(a^2)$ corrections to the effective action are full $SO(4)$ -invariant curvature scalars (no cubic anisotropy until $O(a^4)$, controlled by the degree-six fundamental invariant of F_4). The one-loop graviton self-energy with first-Brillouin-zone cutoff $|\vec{q}| < \pi/a$ is finite at each external momentum, with the standard $O(M_P^4)$, $O(M_P^2 k^2)$, and $O(k^4 \log)$ pieces of perturbative quantum gravity. The construction is linearized at the loop level. Full nonperturbative quantum gravity, the cosmological-constant problem, and resolution of black-hole singularities remain open.

1 Introduction

1.1 The SSMTheory program

The SSMTheory program treats the Face-Centered Cubic (FCC) lattice as the substrate of three-dimensional physical space [1] and the D_4 root lattice as physical four-dimensional space-time [3]. The matter paper [1] establishes that trapped tetrahedral defects in the FCC lattice carry quark quantum numbers and computes the proton-to-electron mass ratio from a single-coupling structural counting. The mass-energy-information paper [2] identifies the five stable mass eigenvalues $\{1, 207, 273, 1836, 1839\}$ with the electron, muon, pion, proton, and neutron via the verification cost of a [[192, 130, 3]] CSS code on the FCC. The CSS code itself is constructed in [4]. The D_4 paper [3] extends the spatial picture to four dimensions, with rank-four plaquette isotropy giving an $SO(4)$ -symmetric Yang-Mills theory at leading lattice order. The linearized gravity paper [5] derives the classical Newton potential from FCC elasticity around tetrahedral defects.

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The present paper completes the gravitational sector at the quantum level: we identify the propagating graviton, derive its quantum properties from the CSS code structure, take the continuum limit to recover nonlinear General Relativity, and show that the lattice ultraviolet cutoff renders one-loop quantum-gravity integrals finite.

1.2 The pure-gauge obstruction and the vison resolution

A natural first attempt to quantize gravity on a lattice is to identify the graviton with phonon excitations of the displacement field $u_\mu(x)$ that describes smooth fluctuations of FCC site positions. This approach fails: a metric perturbation of the form $h_{\mu\nu} = \partial_\mu u_\nu + \partial_\nu u_\mu$ is pure gauge under the linearized diffeomorphism symmetry $h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$ of GR, with gauge parameter $\xi_\mu = -u_\mu$ removing it entirely. Phonon modes of u_μ therefore lie entirely within the gauge-equivalence class and carry no physical content as graviton excitations. The linearized gravity paper [5] accommodates this by treating only the static Kelvin response sourced by lattice defects, where u_μ acquires gauge-invariant content via boundary conditions at defects.

The present paper resolves the obstruction by identifying gravitons not with phonons but with *visons*: localized m-type excitations of the $[[192, 130, 3]]$ CSS code. A vison is created by ceasing to measure one octahedral-void X -stabilizer of the code; it carries a non-trivial topological charge that cannot be removed by any redefinition of u_μ . Coherent plane-wave superpositions of visons produce propagating metric perturbations $h_{\mu\nu}(x)$ that lie outside the pure-gauge equivalence class.

1.3 Main results

We establish five results, no parameter tuned beyond the single identification of the FCC lattice spacing with the Planck length:

Architectural uniqueness (Section 3). Among local discrete excitations of the FCC stabilizer code, the vison is the unique candidate satisfying all five graviton axioms (Theorem 1).

Discrete graviton (Sections 4–6). A single vison produces Regge deficit $\arccos(-1/3)$ at each of the 12 edges of the surrounding octahedron (Proposition 2). Coherent vison waves obey $\omega = c|\vec{k}|$ with masslessness (Proposition 4) and two transverse-traceless spin-2 polarizations (Proposition 5), via the Roček-Williams theorem [8].

Newton’s constant (Section 5). Bekenstein-Hawking entropy matching applied to the 2D sheet-plaquette area $L_0^2 = a^2/2$ fixes $G = a^2/(8 \ln 2)$.

Continuum limit (Section 8). The Cheeger-Müller-Schrader theorem applied to the 4D simplicial structure of D_4 gives convergence of the discrete Regge action to the continuum Einstein-Hilbert action (Theorem 7). The rank-four bond-tensor isotropy of D_4 (Theorem 8) ensures that the leading $O(a^2)$ corrections to the effective action are full $SO(4)$ -invariant curvature scalars (Theorem 9), with the first cubic-anisotropic correction at $O(a^4)$ via the degree-six fundamental invariant of F_4 .

One-loop finiteness (Section 9). The one-loop graviton self-energy on D_4 , with momentum integration restricted to the first Brillouin zone $|\vec{q}| < \pi/a$, is finite at each external momentum (Proposition 10).

1.4 Scope and limitations

This paper is explicitly *linearized at the loop level and classical at the action level*. We do *not* claim:

- All-loop renormalization or a UV completion of perturbative quantum gravity.
- A solution to the cosmological-constant hierarchy. The bare $O(M_P^4)$ vacuum energy density emerges naturally from the lattice cutoff, as in any cutoff-regulated quantum field theory; this reproduces the standard QFT expectation but does not explain the 120-order-of-magnitude smallness of the observed value.
- A resolution of the black-hole information paradox or strong-field interior structure.
- A nonperturbative or background-independent formulation of quantum gravity.
- Concrete predictions distinguishing the D_4 continuum theory from standard GR at currently accessible energies, beyond the $O((E/M_P)^4)$ Lorentz-violation signatures of [3, 5].

What is delivered is a self-contained linearized and one-loop perturbative quantum gravity theory on D_4 , with the discrete-to-continuum bridge and finite one-loop corrections rigorously established. Full nonperturbative quantum gravity remains an open research program.

2 The FCC Stabilizer Code Vacuum

2.1 FCC geometry and structure tensor

The FCC lattice with cubic cell parameter a has nearest-neighbor bond length $L_0 = a/\sqrt{2}$ and $K = 12$ nearest neighbors at the unit vectors

$$\hat{n}_j \in \frac{1}{\sqrt{2}}\{\pm\hat{e}_i \pm \hat{e}_k : i \neq k, \{i, k\} \subset \{x, y, z\}\}. \quad (1)$$

The rank-two structure tensor is

$$S_{\mu\nu} = \sum_{j=1}^{12} \hat{n}_\mu^j \hat{n}_\nu^j = 4 \delta_{\mu\nu}, \quad (2)$$

the maximal eigenvalue for any 3D Bravais lattice and the structural reason for the isotropic long-wavelength response established in [1, 5]. The Delaunay decomposition of FCC fills 3D space with regular tetrahedra (edge L_0 , dihedral $\arccos(1/3)$) and regular octahedra (edge L_0 , dihedral $\arccos(-1/3)$). Each edge is shared by exactly 2 tetrahedra and 2 octahedra, and the dihedral angles satisfy

$$2 \arccos\left(\frac{1}{3}\right) + 2 \arccos\left(-\frac{1}{3}\right) = 2\pi, \quad (3)$$

so the Regge deficit at every interior edge of the pristine FCC lattice vanishes. The vacuum is exactly flat.

2.2 The $[[192, 130, 3]]$ CSS code

The CSS code of [4] places one physical qubit on each edge of the $L = 4$ FCC lattice wrapped on a 3-torus, with two stabilizer types:

$$Z\text{-check at vertex } v : [H_Z]_{v,e} = 1 \iff v \text{ is an endpoint of } e, \quad (4)$$

$$X\text{-check at Ovoid } o : [H_X]_{o,e} = 1 \iff \text{both endpoints of } e \text{ bound void } o. \quad (5)$$

Both checks have weight 12 (each FCC vertex has 12 incident edges; each octahedral void is bounded by 12 edges). CSS validity $H_X H_Z^T = 0 \pmod{2}$ holds because each vertex of an

octahedron has exactly 4 edges within the octahedron—an even number. The code has $n = 192$ edges, $k = 130$ logical qubits (encoding rate 67.7%), and distance $d = 3$.

The vacuum is the simultaneous $+1$ eigenstate of all stabilizers. Local excitations of two types create -1 stabilizer eigenvalues:

- *e-type excitation* (Z -error syndrome): flips an edge qubit so that two adjacent Z -checks register -1 . In the SSM realization this corresponds to a Tvoid insertion at the centroid of a tetrahedral void (Section 7).
- *m-type excitation* (X -error syndrome / removed stabilizer): one octahedral-void X -check is ceased to be measured, freeing one logical qubit ($\Delta k = +1$, $\Delta S = \ln 2$). We call this object the *vison* (Section 4).

2.3 FCC as spatial slice of D_4 : worldlines in spacetime

The CSS code is defined on the 3D FCC lattice, identified in the SSM framework with a spatial time-slice of the 4D D_4 root lattice [3]. The slicing relation is

$$D_4 \cap \{x_4 = \text{const}\} = \text{FCC}. \quad (6)$$

Time evolution of the stabilizer code corresponds to translation along the x_4 axis of D_4 , with each instantaneous time slice carrying a copy of the FCC CSS code and adjacent slices coupled by the D_4 cross-slice bonds.

In this 4D picture the matter and gravitational excitations have natural worldline extensions: a Tvoid insertion at FCC position $\vec{r}$ traces out a quark worldline $\{(\vec{r}, x_4) : x_4 \in \mathbb{R}\}$ in D_4 ; an Ovoid removal at FCC position $\vec{R}_o$ traces out a vison worldline, with the 12-edge spatial Regge deficit (Proposition 2) as the spatial cross-section of a 4D defect structure. The graviton wave is the long-wavelength 4D propagation of coherent vison-worldline superpositions through D_4 ; the Roček-Williams theorem [8] applied to the 4D simplicial structure of D_4 gives the dispersion relation, and the rank-four bond-tensor isotropy of D_4 (Theorem 8 below) ensures Lorentz-invariant propagation at leading lattice order.

3 Architectural Choice: Graviton Axioms and Candidate Enumeration

3.1 Five axioms defining a graviton

The graviton is the quantum of the gravitational field in the linearized approximation around a flat background. The defining properties of this excitation are encoded in five axioms, each drawn from the textbook definition of GR’s massless spin-2 mode:

Axiom 1 (Massless dispersion). *The graviton dispersion satisfies $\omega(\vec{k}) = c|\vec{k}|$ at leading order, with no mass gap: $\omega(\vec{k} = 0) = 0$. A mass term $m_g^2 h_{\mu\nu} h^{\mu\nu}$ is forbidden by linearized diffeomorphism invariance (Fierz-Pauli ghost-freedom [9]).*

Axiom 2 (Spin 2). *The graviton transforms in the helicity- ± 2 representation of the little group $\text{ISO}(2)$ of massless particles in 4D Minkowski space. Starting from 10 independent components of $h_{\mu\nu}$ (symmetric 4×4), linearized diffeomorphism gauge symmetry removes 4 and Bianchi-identity constraints remove 4 more, leaving 2 transverse-traceless (TT) polarizations.*

Axiom 3 (Gauge-invariant physical content). *Linearized GR has the gauge symmetry $h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$. Configurations of the form $h_{\mu\nu} = \partial_\mu u_\nu + \partial_\nu u_\mu$ for smooth u_μ are pure gauge and carry no physical content; a genuine graviton excitation must lie outside this gauge-equivalence class.*

Axiom 4 (Universal coupling to matter). *General Relativity couples to matter via $S_{\text{int}} = -\frac{\kappa}{2} \int d^4x h_{\mu\nu} T^{\mu\nu}$ with a single coupling $\kappa = \sqrt{16\pi G}$ independent of matter species. The equivalence principle requires that all matter types experience identical gravitational coupling.*

Axiom 5 (Continuum limit reproduces linearized Einstein-Hilbert). *The discrete kinetic action, expanded around flat space to quadratic order in $h_{\mu\nu}$, must equal the linearized Einstein-Hilbert action,*

$$S_{\text{EH}}^{(2)} = \frac{1}{16\pi G} \int d^4x \left[-\frac{1}{2}(\partial_\lambda \bar{h}_{\mu\nu})^2 + \frac{1}{4}(\partial_\lambda \bar{h})^2 \right],$$

with $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$. Newton's law $V(r) = -Gm_1m_2/r$ follows from this axiom combined with Axiom 4.

3.2 Candidate enumeration

Local discrete excitations of the FCC stabilizer code on D_4 fall into seven structurally distinct classes:

- A. **Phonons:** smooth displacements $u_\mu(x)$ of FCC site positions.
- B. **Tvoid insertions** (e-type): extra nodes at tetrahedral-void centroids, with four compressed bonds (Section 7).
- C. **Visons** (m-type, Ovoid X -stabilizer removal): ceasing to measure one octahedral-void stabilizer (Section 4).
- D. **Vertex Z -stabilizer removal:** the CSS dual of (C) under the automorphism $X \leftrightarrow Z$.
- E. **Line / dislocation defects:** 1D lattice singularities (Burgers vectors), the discrete analog of Kleinert dislocations [19].
- F. **Plebanski 2-form / paired plaquette excitations:** gauge-theoretic gravitational degrees of freedom from 2-form fields.
- G. **Composite excitations:** bound states of two or more elementary defects (e.g., e - m pairs, m - m pairs).

3.3 Evaluation against axioms

Each candidate is evaluated against the five axioms in Table 1:

Candidate	Ax. 1 Massless	Ax. 2 Spin 2	Ax. 3 Gauge-inv.	Ax. 4 Universal	Ax. 5 Cont. limit	Verdict
A. Phonons (u_μ)	✓	× (spin 1)	× (pure gauge)	✓	×	Reject
B. Tvoid (e-type, matter)	× (massive)	×	✓	—	—	Matter
C. Vison (m-type)	✓	✓	✓	✓	✓	Accept
D. Vertex Z -stab removal	✓	✓	✓	✓	✓	Dual of C
E. Line / dislocation defects	×	× (torsion)	✓	✓	×	Reject
F. Plebanski 2-form	?	✓	✓	?	✓	Open
G. Composite excitations	—	—	—	—	—	Not single-graviton

Table 1: Evaluation of graviton candidates against the five axioms of Section 3.1. Only the vison (C) and its CSS dual (D) pass all axioms among local discrete excitations of the code. Plebanski-style 2-form formulations (F) are a parallel research program not pursued here. Composite excitations (G) are not single-graviton candidates by construction. Tvoid insertions (B) are matter, not gravity.

Phonons (A). A smooth displacement $u_\mu(x)$ is a 4-vector field, so its phonon modes transform as spin 1, failing Axiom 2. The metric perturbation $h_{\mu\nu} = \partial_\mu u_\nu + \partial_\nu u_\mu$ is pure gauge, failing Axiom 3: setting $\xi_\mu = -u_\mu$ removes h entirely. The TT graviton polarizations are not in the image of $u \mapsto \partial u + \partial u$.

Tvoid insertions (B). Matter excitations, with rest mass $m = 2J(1 - \sqrt{6}/4)^2 \neq 0$ (Section 7), failing Axiom 1. Not a candidate for the gravitational sector.

Vertex Z -stab removal (D). The CSS code has an exact automorphism $X \leftrightarrow Z$ exchanging H_X and H_Z . Under this automorphism, vertex Z -stabilizer removal maps to Ovoid X -stabilizer removal: the two excitations are physically equivalent up to relabeling. Candidate D is therefore the CSS dual of candidate C, not a distinct alternative.

Line defects (E). Lattice dislocations produce torsion (a rank-3 tensor with antisymmetric pair) rather than curvature, failing Axiom 5 for standard GR. They give rise to Einstein-Cartan rather than Einstein-Hilbert gravity, a distinct theory.

Plebanski 2-forms (F). The 2D plaquette structure of D_4 would naturally accommodate a Plebanski-type 2-form formulation of gravity, in which the gravitational degrees of freedom emerge from paired plaquette holonomies rather than removed-stabilizer excitations. Whether this construction satisfies Axioms 1 and 4 is an open question whose detailed analysis lies outside the scope of this paper.

Composite excitations (G). Bound states of two or more elementary defects are not single-graviton candidates by construction: $e + m$ pairs describe matter propagation in a curved background (the standard matter-gravity interaction), and $m + m$ pairs describe two-graviton states.

Theorem 1 (Vison uniqueness among local discrete excitations). *Among the seven structurally distinct local discrete excitations of the $[[192, 130, 3]]$ CSS code on D_4 , the vison (Ovoid X -stabilizer removal) and its CSS dual (vertex Z -stabilizer removal) are the unique candidates satisfying all five graviton axioms (Axioms 1–5). Plebanski-style 2-form excitations remain an open alternative architecture not pursued here.*

4 Visions: Curvature from Ovoid Removal

4.1 Vison creation and the freed logical qubit

To create a vison at octahedral-void position $\vec{R}_o$, we cease to measure the X -check at o but leave all other stabilizers active. The resulting code has one fewer constraint, so its logical-qubit count increases by $\Delta k = +1$. The corresponding entanglement entropy change is

$$\Delta S = \ln 2 \quad \text{per removed stabilizer,} \quad (7)$$

since a freed logical qubit contributes one bit of entropy to the code state. This ΔS is the fundamental quantum of entropy in the SSM gravitational sector and will fix Newton’s constant in Section 5.

4.2 Regge curvature: deficit at 12 edges

Proposition 2 (Vison curvature). *Removing the X -stabilizer at octahedral void o produces Regge deficit angle*

$$\delta_e = \arccos\left(-\frac{1}{3}\right) \approx 109.47^\circ$$

at each of the 12 edges of the octahedron bounding o , and $\delta_e = 0$ at all other edges of the FCC lattice.

Proof. Each FCC edge in the pristine lattice is shared by exactly 2 tetrahedra and 2 octahedra, with total dihedral angle

$$\sum_{\sigma \supset e} \theta_{\sigma,e} = 2 \arccos \frac{1}{3} + 2 \arccos\left(-\frac{1}{3}\right) = 2\pi \quad (8)$$

by (3). The discontinuation of the X -measurement at o removes the octahedron o from the active cell complex (its interior is no longer constrained by the stabilizer). The 12 edges of o each lose one $\arccos(-1/3)$ contribution from the now-removed octahedral cell, leaving total dihedral

$$\sum_{\sigma' \supset e, \sigma' \neq o} \theta_{\sigma',e} = 2 \arccos \frac{1}{3} + \arccos\left(-\frac{1}{3}\right), \quad (9)$$

giving Regge deficit

$$\begin{aligned} \delta_e &= 2\pi - \left[2 \arccos \frac{1}{3} + \arccos\left(-\frac{1}{3}\right)\right] \\ &= \left[2 \arccos \frac{1}{3} + 2 \arccos\left(-\frac{1}{3}\right)\right] - \left[2 \arccos \frac{1}{3} + \arccos\left(-\frac{1}{3}\right)\right] \\ &= \arccos\left(-\frac{1}{3}\right). \end{aligned} \quad (10)$$

Edges not bounding o remain shared by 2 tetrahedra and 2 octahedra, so $\delta_e = 0$ there. $\square$

The 12 affected edges are distributed equally across the three coordinate planes: 4 edges in each of the three triad sheets [4] containing o . A single vison therefore couples all three spatial directions, consistent with its identification as a scalar quantum of curvature.

4.3 The vison carries gauge-invariant physical content (Axiom 3)

A vison is not a smooth displacement field but a discrete topological excitation of the CSS code, characterized by a non-trivial X -syndrome that cannot be undone by any local unitary operation on the qubits. The Regge deficit of Proposition 2 is a 2π multi-valued angle defect of the connection, which cannot be removed by any redefinition of u_μ : it lies outside the pure-gauge equivalence class. The vison thus satisfies Axiom 3, and its coherent waves carry physical content in the linearized GR sense.

5 Newton's Constant from Entropy Matching

5.1 The 2D sheet plaquette

The CSS code of [4] has a layered structure: each triad sheet S_{xy}, S_{xz}, S_{yz} is a 2D toric code on a $K = 4$ rotated square lattice with plaquettes of side $L_0 = a/\sqrt{2}$. The 2D sheet plaquette is a square of side L_0 with area

$$A_{\text{plaq}} = L_0^2 = \frac{a^2}{2}. \quad (11)$$

This is the geometric primitive of the layered X -stabilizer structure: each Ovoid X -check in the 3D FCC code is built from the four edges of one sheet plaquette in each of the three triad

sheets it touches. Removing the Ovoid corresponds to removing one such sheet plaquette’s measurement constraint.

We emphasize that A_{plaq} is *not* the area of a triangular face of the 3D FCC Delaunay complex (which would be $\sqrt{3}a^2/8 \approx 0.217a^2$). The 2D sheet plaquette is a distinct geometric object inherited from the 2D toric-code layers of [4].

5.2 Bekenstein-Hawking matching

Applying the Bekenstein-Hawking formula $\Delta S = \Delta A/(4G)$ [10] to a single removed Ovoid stabilizer with $\Delta S = \ln 2$ (eq. (7)) and $\Delta A = A_{\text{plaq}} = a^2/2$ (eq. (11)):

$$\boxed{G = \frac{a^2}{8 \ln 2}}. \quad (12)$$

Equivalently $a = \sqrt{8 \ln 2} \ell_P \approx 2.355 \ell_P$ when $G = \ell_P^2$. The algebraic identity $L_0^2 = 4G \ln 2$ follows immediately. By linearity, applying $\Delta S = \Delta A/(4G)$ to a horizon of area A comprising $N = A/L_0^2$ removed Ovoid stabilizers gives the standard Bekenstein-Hawking entropy $S_{\text{BH}} = A/(4G)$ as a direct consequence, with each bit of horizon entropy corresponding to one removed stabilizer.

5.3 One parameter, not zero

Equation (12) is not a derivation of Newton’s constant from no input. It uses one structural input—the 2D sheet-plaquette area $A_{\text{plaq}} = L_0^2$ inherited from the CSS code of [4]—and the standard Bekenstein-Hawking relation. All other parameters of the theory (c , J , κ , matter masses) follow from (12) and FCC geometry by algebraic identities. Given that single input, no further tuning is possible.

6 The Graviton as Coherent Vison Wave

6.1 From vison amplitudes to metric perturbations

Let $\hat{a}_o^\dagger, \hat{a}_o$ be creation and annihilation operators for the vison at octahedral void position $\vec{R}_o$, satisfying the bosonic algebra $[\hat{a}_o, \hat{a}_{o'}^\dagger] = \delta_{oo'}$ (the visons are bosonic since the underlying X -syndromes commute). A coherent vison state

$$|\phi\rangle \propto \exp\left(\sum_o \phi_o \hat{a}_o^\dagger\right) |\Omega\rangle \quad (13)$$

satisfies the eigenvalue equation $\hat{a}_o |\phi\rangle = \phi_o |\phi\rangle$, with $\phi_o \in \mathbb{C}$ the vison amplitude at $\vec{R}_o$.

For a plane-wave amplitude $\phi_o = h e^{i\vec{k} \cdot \vec{R}_o}$, the expected Regge deficit at edge e is, to leading order in h ,

$$\langle \delta_e \rangle = h f_e(\vec{k}) e^{i\vec{k} \cdot \vec{x}_e} + \text{c.c.}, \quad (14)$$

where $f_e(\vec{k})$ is a geometric form factor encoding the projection of the vison curvature pattern (Proposition 2) onto edge e . In the long-wavelength limit $|\vec{k}|a \ll 1$, this collective pattern is exactly the linearized metric perturbation

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x), \quad h_{\mu\nu}(x) = h e_{\mu\nu}(\hat{k}) e^{i\vec{k} \cdot x}, \quad (15)$$

with polarization tensor $e_{\mu\nu}(\hat{k})$ fixed by the vison geometry (Section 6.5).

6.2 Linearized Regge = linearized Einstein-Hilbert

The expectation value of the SSM stabilizer Hamiltonian $H = -J \sum_s A_s$ on a vison coherent state, expanded to quadratic order in h , defines the vison kinetic action $S_{\text{vison}}^{(2)}[h]$. We require this action's continuum limit.

Theorem 3 (Roček-Williams [8]). *On a flat simplicial lattice in d dimensions, the linearized Regge action obtained by expanding $\frac{1}{16\pi G} \sum_e l_e \delta_e$ around the flat-edge configuration is identical (up to surface terms) to the linearized Einstein-Hilbert action,*

$$S_{\text{Regge}}^{(2)} = \frac{1}{16\pi G} \int d^d x \left[-\frac{1}{2} (\partial_\lambda \bar{h}_{\mu\nu})^2 + \frac{1}{4} (\partial_\lambda \bar{h})^2 \right], \quad (16)$$

with $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h$. The discrete gauge invariance (vertex displacements $x_v \rightarrow x_v + \xi_v$) maps to linearized diffeomorphism invariance ($h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$).

Applying Theorem 3 to the 4D simplicial structure of D_4 with G fixed by (12), the vison kinetic action equals the standard Fierz-Pauli action for a massless symmetric rank-two tensor. This verifies Axiom 5 for the vison architecture.

6.3 Dispersion $\omega = c|\vec{k}|$ (Axiom 1)

The equation of motion from (16) in harmonic gauge $\partial^\mu \bar{h}_{\mu\nu} = 0$ is

$$\square \bar{h}_{\mu\nu} = 0, \quad \square = -c^{-2} \partial_t^2 + \nabla^2. \quad (17)$$

Substituting the plane wave (15) gives

$$\omega = c|\vec{k}|, \quad (18)$$

verifying Axiom 1. The speed c is fixed by the FCC structure tensor (2): by [4, Lemma 11.1], all O_h -invariant rank- ≤ 3 tensors are $SO(3)$ -invariant, so the dispersion (18) is isotropic in all spatial directions at leading lattice order. Corrections begin at $O((|\vec{k}|a)^2)$ and remain $SO(4)$ -invariant at $O(a^2)$ by the rank-four isotropy theorem (Theorem 8 below).

6.4 Masslessness

Proposition 4 (Graviton masslessness in the FCC vacuum). *In the pristine FCC vacuum (no matter sources), the coherent vison wave (15) satisfies $\omega(\vec{k} = 0) = 0$.*

Proof. Two independent arguments, both holding in the pristine vacuum:

Argument 1 (Regge diffeomorphism invariance). The linearized Regge action of Theorem 3 is invariant under vertex displacements $x_v \rightarrow x_v + \xi_v$. By the Schläfli identity $\partial S_{\text{Regge}} / \partial l_e = \delta_e$, and since $\delta_e = 0$ in the pristine FCC vacuum by (3), infinitesimal vertex displacements cost zero action. This gauge invariance forbids a mass term $m_g^2 \bar{h}_{\mu\nu} \bar{h}^{\mu\nu}$, which would explicitly break the invariance.

Argument 2 (vison spectrum translation symmetry). In the pristine FCC vacuum, every octahedral void is geometrically equivalent (the FCC lattice is a single $W(D_3) \rtimes T^3$ orbit on the Ovoid sites). The vison energy is therefore the same at every Ovoid position, so the uniform ($\vec{k} = 0$) vison mode is a zero mode of the lattice Hamiltonian. A finite mass would require a gap at $\vec{k} = 0$; no such gap exists. $\square$

When matter sources (Tvoids) are present, both arguments are modified: vertex positions are partially constrained by Tvoid bonds, and the vison spectrum is no longer translation-invariant. The resulting massless graviton couples to matter via the standard linearized stress-energy coupling (Section 7), reproducing Newton's law.

6.5 Spin 2: two transverse-traceless polarizations (Axiom 2)

Proposition 5 (Two helicity- ± 2 polarizations). *The coherent vison wave (15) has exactly two physical degrees of freedom, transforming as helicity $\lambda = \pm 2$ under rotations about the propagation direction.*

Proof. Starting from the 10 independent components of $h_{\mu\nu}$ (symmetric 4×4): the linearized diffeomorphism gauge symmetry $h \rightarrow h + \partial\xi + \partial\xi$ removes 4 components (one per spacetime direction); the Bianchi-identity constraints $\partial^\mu \bar{h}_{\mu\nu} = 0$ remove 4 more on-shell. This leaves $10 - 4 - 4 = 2$ physical degrees of freedom, identical to continuum linearized GR [9]. The Roček-Williams algebra of discrete diffeomorphisms matches the continuum gauge algebra by Theorem 3, so no additional lattice modes appear.

For propagation $\vec{k} = k\hat{z}$, the surviving components are $h_+ = \frac{1}{\sqrt{2}}(h_{xx} - h_{yy})$ and $h_\times = h_{xy}$. We verify their helicity assignment by decomposing $h_{\mu\nu}$ under the $C_4 \subset O_h$ rotation by angle φ about $\hat{z}$:

$$\text{helicity } 0 : \quad h_{zz}, h_{xx} + h_{yy} \rightarrow e^{0i\varphi}(\cdot), \quad (19)$$

$$\text{helicity } \pm 1 : \quad h_{xz} \pm ih_{yz} \rightarrow e^{\pm i\varphi}(\cdot), \quad (20)$$

$$\text{helicity } \pm 2 : \quad (h_{xx} - h_{yy}) \pm 2ih_{xy} \rightarrow e^{\pm 2i\varphi}(\cdot). \quad (21)$$

The transversality condition $k^\mu \bar{h}_{\mu\nu} = 0$ eliminates $h_{z\nu}$ for all ν , removing the helicity-0 component h_{zz} and both helicity- ± 1 components. Tracelessness $\bar{h}^\mu{}_\mu = 0$ removes the remaining helicity-0 combination $h_{xx} + h_{yy}$. The surviving modes $h_+, h_\times$ transform as $e^{\pm 2i\varphi}$, giving helicity $\lambda = \pm 2$. No ghost modes (helicity-0 scalar gravity, helicity- ± 1 vector gravity) survive the constraint structure. $\square$

7 Matter Sector and the Equivalence Principle

7.1 Tvoid geometry and bond compression

A tetrahedral void in the FCC unit cell is bounded by four FCC sites A, B, C, D forming a regular tetrahedron of edge L_0 . The centroid-to-vertex distance is the circumradius of the regular tetrahedron,

$$r_0 = \frac{\sqrt{6}}{4} L_0 \approx 0.6124 L_0. \quad (22)$$

Inserting a node at this centroid and bonding it to A, B, C, D creates four bonds whose natural length is L_0 (the Bell-pair spacing of [4]) but whose geometrically forced length is r_0 . The fractional compression is

$$\frac{\Delta L}{L_0} = 1 - \frac{\sqrt{6}}{4} \approx 0.3876 \quad (38.76\%), \quad (23)$$

an exact FCC geometric number with no free parameter.

7.2 Mass from bond compression

Expanding the Wilson plaquette action $J(1 - \cos \theta)$ to quadratic order in the bond-strain angle $\theta = \delta l / L_0$ gives spring constant $k_{\text{spring}} = J / L_0^2$. The four compressed bonds of the Tvoid each contribute energy $\frac{1}{2} k_{\text{spring}} (\Delta L)^2$, summing to

$$m = 4 \cdot \frac{1}{2} \cdot \frac{J}{L_0^2} \cdot (L_0 - r_0)^2 = 2J \left(1 - \frac{\sqrt{6}}{4}\right)^2. \quad (24)$$

The stabilizer coupling J is fixed by matching to Newton's constant via $J = G / (32\pi L_0^3)$, derived from the Roček-Williams normalization of the FCC stiffness matrix [5, 8]. Substituting (12), this gives $J \approx 5.07 \times 10^{-3} m_{\text{PC}}^2$ in Planck units.

7.3 Discrete Birkhoff: matter is geometrically neutral at the lattice scale

Theorem 6 (Discrete Birkhoff for the FCC lattice). *Inserting a Tvoid node at the centroid of a regular tetrahedron $ABCD$, with the Tvoid bonds forced to length r_0 , produces zero Regge deficit angle at every edge of the lattice.*

Proof. Regge curvature depends on the geometric dihedral angles at each edge, which in turn depend only on the *positions* of the surrounding vertices, not on the natural lengths of the bonds connecting them. The bond compression (23) contributes to the elastic energy (24) but not to the dihedral geometry.

Original tetrahedron edges (e.g., AB). Each original edge is now bounded by two new sub-tetrahedra ($\{V_T, A, B, C\}$ and $\{V_T, A, B, D\}$ in place of the original $ABCD$, with V_T the Tvoid centroid) plus the two unchanged octahedra. By direct computation, the two sub-tetrahedron dihedrals at AB each equal $\arccos(2/\sqrt{6})$. Using the identity $\cos(2 \arccos(2/\sqrt{6})) = 2(4/6) - 1 = 1/3$, we obtain

$$2 \arccos\left(\frac{2}{\sqrt{6}}\right) = \arccos\frac{1}{3}, \quad (25)$$

exactly the original tetrahedron's contribution. The dihedral sum at AB is therefore unchanged at 2π , and $\delta_{AB} = 0$.

New Tvoid edges (e.g., V_TA). Three sub-tetrahedra share V_TA . By the three-fold symmetry of $\{B, C, D\}$ about the axis through V_T and A , each contributes the same dihedral angle. Direct computation gives $120^\circ = 2\pi/3$ each, summing to 2π , hence $\delta_{V_TA} = 0$.

All other FCC edges. Original FCC edges not bounding the affected tetrahedron are surrounded by unchanged cells, with dihedral sum 2π by (3), hence $\delta = 0$. $\square$

Theorem 6 states that SSM matter does not curve the discrete FCC lattice. Curvature emerges only in the continuum limit through the elastic response of the surrounding lattice to the bond prestress—the Kelvin solution of [5]—which sources $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ at long wavelengths.

7.4 Universal coupling from braiding phase (Axiom 4)

In any CSS stabilizer code, the braiding phase acquired when an e-type excitation winds around an m-type excitation is a topological invariant: (-1) per unit linking number, independent of the internal structure of either excitation. This is a consequence of the $\mathbb{F}_2$ structure of the stabilizer group.

In the SSM gravitational context, this universal braiding phase is the microscopic origin of the gravitational equivalence principle. Transporting a Tvoid matter particle (of any internal structure) along a closed loop γ around a vison worldline produces a path-ordered code-Hilbert-space holonomy

$$\mathcal{A}_\gamma = (-1)^{\text{link}(\gamma, \text{vison})}. \quad (26)$$

In the continuum limit, this discrete topological selection rule becomes the statement that the gravitational Berry phase acquired by a matter particle in a curved background is the same functional of $h_{\mu\nu}$ for all matter species. The standard linearized matter-gravity coupling

$$S_{\text{int}} = -\frac{\kappa}{2} \int d^4x h_{\mu\nu} T^{\mu\nu}, \quad \kappa = \sqrt{16\pi G}, \quad (27)$$

is therefore universal: the coefficient κ is the same for all Tvoid species, verifying Axiom 4.

7.5 Tree-level Newton's law

Tree-level graviton exchange between two non-relativistic Tvoid masses m_1, m_2 via (27) gives the standard scattering amplitude

$$\mathcal{M} = -\frac{16\pi G m_1 m_2}{|\vec{k}|^2}, \quad (28)$$

whose Fourier transform is the Newton potential $V(r) = -G m_1 m_2 / r$. The $1/|\vec{k}|^2$ propagator pole is the massless graviton propagator from (17).

8 Continuum Limit: Cheeger-Müller-Schrader on D_4

8.1 Discrete Regge action on D_4

For a simplicial 4-complex K with edge lengths $\{l_e\}$, the Regge action [6] in 4D is

$$S_{\text{Regge}}[K, \{l_e\}] = \frac{1}{16\pi G} \sum_{h \in \mathcal{H}} A_h \delta_h, \quad (29)$$

where $\mathcal{H}$ is the set of hinges (2-simplices in 4D), A_h is the area of hinge h , and $\delta_h = 2\pi - \sum_{\sigma \supset h} \theta_{\sigma, h}$ is the Regge deficit angle at h .

8.2 The CMS theorem and its application to D_4

Theorem 7 (Cheeger-Müller-Schrader [7]). *Let (M, g) be a compact Riemannian 4-manifold and $\{K_n\}_{n \in \mathbb{N}}$ a sequence of simplicial approximations with edge lengths $\{l_e^{(n)}\}$ and mesh size $\text{mesh}(K_n) \rightarrow 0$ as $n \rightarrow \infty$. Assume the simplices are non-degenerate (all dihedral angles bounded away from 0 and π). Then*

$$\lim_{n \rightarrow \infty} \frac{1}{16\pi G} \sum_{h \in K_n} A_h^{(n)} \delta_h^{(n)} = \frac{1}{16\pi G} \int_M R \sqrt{-g} d^4x, \quad (30)$$

with convergence in the sense of distributions. The convergence rate is $O(\text{mesh}(K_n)^2)$ for smooth g .

The D_4 lattice with spacing a provides a uniform simplicial approximation to flat spacetime $\mathbb{R}^4$ with mesh size $a\sqrt{2}$ (the bond length $L_0 = a\sqrt{2}$). Non-degeneracy is automatic: all simplex dihedral angles on D_4 are bounded by those of the regular 4-simplex of side L_0 , which lies in the interior of $(0, \pi)$. Theorem 7 therefore applies directly:

$$\boxed{\frac{1}{16\pi G} \sum_{h \in K_{D_4}} A_h \delta_h \xrightarrow{a \rightarrow 0} \frac{1}{16\pi G} \int_M R \sqrt{-g} d^4x.} \quad (31)$$

Combined with the matter coupling derived in Section 7, the full continuum theory on D_4 is

$$S = \frac{1}{16\pi G} \int d^4x \sqrt{-g} R + S_{\text{matter}}[g, \text{Tvoids}], \quad (32)$$

which is classical General Relativity coupled to matter, with $G = a^2/(8 \ln 2)$ fixed by (12).

8.3 Rank-four isotropy of D_4

We now prove the central algebraic theorem controlling the lattice corrections to the continuum limit. The 24 nearest neighbors of the origin in D_4 are

$$\mathcal{N}_{D_4} = \{\pm e_\mu \pm e_\nu : 0 \leq \mu < \nu \leq 3\}, \quad (33)$$

with bond length $\sqrt{2}$ in lattice units. The rank-four bond tensor is

$$T_{\mu\nu\rho\sigma}^{(4)} = \sum_{n \in \mathcal{N}_{D_4}} n_\mu n_\nu n_\rho n_\sigma, \quad (34)$$

and the fully-symmetric isotropic rank-four tensor in 4D (unique up to scale) is

$$S_{\mu\nu\rho\sigma}^{(4)} = \delta_{\mu\nu} \delta_{\rho\sigma} + \delta_{\mu\rho} \delta_{\nu\sigma} + \delta_{\mu\sigma} \delta_{\nu\rho}. \quad (35)$$

Theorem 8 (Rank-four isotropy of D_4). *The rank-four bond tensor on D_4 is proportional to the isotropic rank-four tensor:*

$$T_{\mu\nu\rho\sigma}^{(4)} = 4 S_{\mu\nu\rho\sigma}^{(4)}. \quad (36)$$

Proof. By direct contraction over the 24 vectors in $\mathcal{N}_{D_4}$. Each $n \in \mathcal{N}_{D_4}$ has exactly two nonzero components, both equal to ± 1 . We evaluate $T_{\mu\nu\rho\sigma}^{(4)}$ in three index-symmetry classes:

Class $\mu = \nu = \rho = \sigma$ (e.g., $\mu = \nu = \rho = \sigma = 0$): Each $n \in \mathcal{N}_{D_4}$ contributes $n_0^4 \in \{0, 1\}$. There are 12 vectors with $n_0 \neq 0$ (the 12 vectors with index 0 paired with any of $\nu = 1, 2, 3$ and signs $\pm$), giving

$$T_{0000}^{(4)} = 12.$$

The isotropic tensor gives $S_{0000}^{(4)} = 3$, so $T_{0000}^{(4)} = 4 \cdot S_{0000}^{(4)}$. ✓

Class $\mu = \nu \neq \rho = \sigma$ (e.g., $\mu = \nu = 0, \rho = \sigma = 1$): Each $n \in \mathcal{N}_{D_4}$ contributes $n_0^2 n_1^2 \in \{0, 1\}$. There are exactly 4 vectors with both $n_0 \neq 0$ and $n_1 \neq 0$ (the 4 sign choices on $\pm e_0 \pm e_1$), giving

$$T_{0011}^{(4)} = 4.$$

The isotropic tensor gives $S_{0011}^{(4)} = \delta_{00}\delta_{11} + \delta_{01}\delta_{01} + \delta_{01}\delta_{01} = 1$, so $T_{0011}^{(4)} = 4 \cdot S_{0011}^{(4)}$. ✓

Class with three or more distinct indices (e.g., $\mu = 0, \nu = 1, \rho = 2, \sigma = 3$): Each $n \in \mathcal{N}_{D_4}$ has only two nonzero components, so $n_0 n_1 n_2 n_3 = 0$ for every n , giving

$$T_{0123}^{(4)} = 0.$$

The isotropic tensor also vanishes: $S_{0123}^{(4)} = 0$ since no two of $\{0, 1, 2, 3\}$ are equal. ✓

All other index combinations reduce to these three classes by the full symmetry of $T^{(4)}$ and $S^{(4)}$ under permutations of (μ, ν, ρ, σ) . Hence $T_{\mu\nu\rho\sigma}^{(4)} = 4 S_{\mu\nu\rho\sigma}^{(4)}$ for all index combinations. □

Theorem 8 is the precise content of the rank-four isotropy claim of [3, 5], now stated self-contained for the present paper without external dependence.

8.4 $SO(4)$ at $O(a^2)$: no cubic-anisotropic curvature correction

For finite a , the discrete Regge action receives lattice corrections relative to the Einstein-Hilbert limit (30). These corrections are organized as a derivative expansion in $a \times (\text{curvature scale})$:

$$S_{\text{eff}}[g] = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[R + a^2 \mathcal{L}_2[R] + a^4 \mathcal{L}_4[R] + \dots \right]. \quad (37)$$

The structure of $\mathcal{L}_n$ is constrained by the F_4 point-group symmetry of the D_4 lattice: any term in $\mathcal{L}_n$ must be F_4 -invariant when written as a contraction of curvature tensors with lattice tensors built from $\mathcal{N}_{D_4}$.

Theorem 9 ($SO(4)$ -isotropy of $O(a^2)$ curvature corrections). *The $O(a^2)$ correction $\mathcal{L}_2[R]$ to the continuum Einstein-Hilbert action on D_4 is a fully $SO(4)$ -invariant combination of curvature scalars,*

$$\mathcal{L}_2[R] = c_1 R^2 + c_2 R_{\mu\nu} R^{\mu\nu} + c_3 R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}, \quad (38)$$

with no cubic-anisotropic correction at this order. The coefficients c_1, c_2, c_3 are pure numbers determined by the D_4 simplicial geometry.

Proof. A generic F_4 -invariant correction at $O(a^2)$ involves two factors of curvature contracted with a rank-four lattice tensor:

$$\mathcal{L}_2 \sim T_{\mu\nu\rho\sigma}^{(4)} R^{\mu\nu\alpha\beta} R^{\rho\sigma}_{\alpha\beta} + (\text{permutations and traces}). \quad (39)$$

By Theorem 8, every appearance of $T_{\mu\nu\rho\sigma}^{(4)}$ is proportional to $S_{\mu\nu\rho\sigma}^{(4)} = \delta_{\mu\nu}\delta_{\rho\sigma} + \delta_{\mu\rho}\delta_{\nu\sigma} + \delta_{\mu\sigma}\delta_{\nu\rho}$. After contracting $S^{(4)}$ with the two curvature tensors, the result is a sum of products of curvature scalars and traces: R^2 , $R_{\mu\nu} R^{\mu\nu}$, and $R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}$, all $SO(4)$ -invariant. No cubic-anisotropic term survives. □

8.5 First anisotropy at $O(a^4)$ via the rank-six tensor

At $O(a^4)$, the relevant lattice tensor is the rank-six bond tensor

$$T_{\mu\nu\rho\sigma\alpha\beta}^{(6)} = \sum_{n \in \mathcal{N}_{D_4}} n_\mu n_\nu n_\rho n_\sigma n_\alpha n_\beta. \quad (40)$$

This tensor is *not* proportional to the fully-isotropic rank-six tensor $S^{(6)}$ on D_4 . The ratios $T^{(6)}/S^{(6)}$ in the three fully-symmetric component classes are

$$\frac{T_{(6,0,0,0)}^{(6)}}{S_{(6,0,0,0)}^{(6)}} = \frac{12}{15} = \frac{4}{5}, \quad \frac{T_{(4,2,0,0)}^{(6)}}{S_{(4,2,0,0)}^{(6)}} = \frac{4}{3}, \quad \frac{T_{(2,2,2,0)}^{(6)}}{S_{(2,2,2,0)}^{(6)}} = \frac{0}{1} = 0, \quad (41)$$

manifestly unequal: $T^{(6)}$ has a non-isotropic (F_4 -invariant but not $SO(4)$ -invariant) component. The structural reason is that the F_4 Weyl group (order 1152, the point group of D_4 [11]) has fundamental invariants of degrees $\{2, 6, 8, 12\}$: a new independent invariant first appears at degree 6, breaking rank-six isotropy.

The first cubic-anisotropic correction to the effective action therefore enters at $O(a^4)$, parametrically suppressed by an additional factor $(a \times \text{curvature scale})^2 \sim (E/M_P)^2$ relative to the leading $SO(4)$ -invariant corrections of Theorem 9.

9 One-Loop Graviton Self-Energy with Lattice Cutoff

9.1 Setup

The linearized graviton on D_4 has the standard Fierz-Pauli propagator and a cubic self-interaction inherited from the nonlinear Einstein-Hilbert action via Theorem 7. The one-loop self-energy is

$$\Pi_{\mu\nu,\rho\sigma}(k) = \frac{\kappa^2}{2} \int_{\text{BZ}_{D_4}} \frac{d^4 q}{(2\pi)^4} \frac{N_{\mu\nu,\rho\sigma}(q, k)}{q^2 (q+k)^2}, \quad (42)$$

where $\kappa = \sqrt{16\pi G}$, the integration is over the first Brillouin zone of D_4 ($|\vec{q}| \lesssim \pi/a$ in any direction), and $N_{\mu\nu,\rho\sigma}(q, k)$ is the polynomial tensor structure from the cubic graviton vertices.

9.2 Finiteness

Proposition 10 (Finiteness of one-loop graviton self-energy on D_4). *The integral (42) is finite for every external momentum k with $|\vec{k}| < \pi/a$. It decomposes as*

$$\Pi_{\mu\nu,\rho\sigma}(k) = \Lambda_{\mu\nu,\rho\sigma}^{(0)} M_P^4 + \Lambda_{\mu\nu,\rho\sigma}^{(2)}(k) M_P^2 + \Lambda_{\mu\nu,\rho\sigma}^{(4)}(k) \log(M_P/|k|) + (\text{finite remainder}), \quad (43)$$

where each $\Lambda^{(n)}$ is a finite tensor function and $M_P^{-1} \equiv a/\pi$ is the lattice ultraviolet scale.

Proof. The integration region BZ_{D_4} is compact, with $|\vec{q}| < \pi/a$ in any direction. The integrand $N_{\mu\nu,\rho\sigma}(q, k)/[q^2(q+k)^2]$ has poles only at $q = 0$ and $q = -k$, both of which are integrable in 4D ($\int d^4 q/q^2 \sim \int q^3 dq/q^2 \sim q^2$ near $q = 0$). The integral is therefore absolutely convergent. The asymptotic decomposition (43) follows from expanding $N_{\mu\nu,\rho\sigma}(q, k)$ in powers of q and k and standard power counting. $\square$

9.3 Physical interpretation

$O(M_P^4)$ piece (cosmological-constant hierarchy). The leading term contributes a constant to the effective potential, $\sim M_P^4 \approx (10^{19} \text{ GeV})^4$. The lattice cutoff renders this term *finite*, in contrast to dimensional regularization (where it appears as a power-law divergence) or

naive Pauli-Villars schemes; the present framework therefore reproduces the standard “natural” size of the vacuum energy expected from any cutoff-regulated quantum field theory [13]. The observed cosmological constant is some 120 orders of magnitude smaller than this natural value, a hierarchy problem common to all known approaches to quantum gravity and effective field theory in curved spacetime. The D_4 construction is fully consistent with this standard picture: it reproduces the expected QFT behavior, including the known unsolved hierarchy. We do not claim to explain why the physical value is small; the discrepancy is inherited unchanged from standard quantum field theory and remains open.

$O(M_P^2 k^2)$ piece (G -renormalization). The $\Lambda^{(2)} M_P^2$ piece has a coefficient proportional to k^2 at leading order (by Lorentz covariance and the Ward identity), corresponding to a finite renormalization of Newton’s constant. The bare $G = a^2/(8 \ln 2)$ from (12) is shifted by this loop correction.

$O(k^4 \log)$ piece (standard counterterms). The $\Lambda^{(4)} \log(M_P/|k|)$ piece corresponds to the standard one-loop R^2 , $R_{\mu\nu} R^{\mu\nu}$, and $R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}$ counterterms of perturbative quantum gravity [12]. On D_4 , these are finite and their coefficients c_1, c_2, c_3 of Theorem 9 are determined by the explicit one-loop integral, which we do not evaluate in closed form here.

9.4 Scope of one-loop result

What is claimed: the one-loop graviton self-energy on D_4 is *finite* at every external momentum below the lattice cutoff. This is a concrete demonstration that the discrete substrate regulates the linearized UV divergences of perturbative quantum gravity. What is *not* claimed: all-loop finiteness (higher-loop integrals have additional momentum integrations, each independently cut off at π/a ; finiteness at each loop order does not imply convergence of the sum); a UV completion of perturbative quantum gravity beyond one loop; a solution to the cosmological-constant problem.

10 Discussion

10.1 Position relative to other quantum-gravity programs

Sakharov-style induced gravity [14, 15]. Derives Newton’s constant from integrating out high-momentum modes of matter fields on a curved background, leaving G as an undetermined proportionality constant. The present D_4 framework fixes G via a single entropy-matching condition (12), and provides the simplicial regularization for the continuum limit via Cheeger-Müller-Schrader.

Loop quantum gravity and spin foams [16, 17]. Share the discrete-curvature/Regge-calculus core but take spin networks as fundamental. The present construction uses the CSS-stabilizer-code structure directly, with the m-type stabilizer excitation (vison) encoding a Regge deficit and bypassing the spin-network coarse-graining step.

Causal dynamical triangulation [18]. Uses dynamical simplicial geometry with weighted sum over triangulations. The present framework keeps the lattice fixed (the D_4 root lattice) and treats the metric perturbation as the dynamical field, with quantization via the CSS-code excitation structure.

Topological-order programs [20]. Demonstrate emergence of gauge bosons and fermions from local entanglement. Extending to emergent spin-2 in 3+1D faces the Weinberg-Witten obstruction and proliferation of lower-spin modes. The present construction circumvents both: spin-2 emerges from the dual sector of the same code that already encodes matter (Axiom 2 verified by Proposition 5), with the rigid O_h crystal symmetry eliminating lower-spin ghosts.

10.2 The Plebanski alternative

An alternative architecture not pursued here is the Plebanski-style 2-form formulation, in which the gravitational degrees of freedom emerge from paired plaquette holonomies rather than removed-stabilizer excitations. The 2D nature of plaquettes in D_4 naturally accommodates this construction. Whether the Plebanski architecture satisfies Axioms 1 and 4 requires detailed analysis we do not undertake here; the present work demonstrates that the vison architecture satisfies all five axioms and reproduces standard GR in the continuum limit.

10.3 Testable consequences and what remains open

The leading lattice corrections of Theorem 9 are full $SO(4)$ scalars and not directly testable from symmetry alone (they appear as standard higher-curvature corrections to GR). The first *anisotropic* corrections at $O((E/M_P)^4)$ predict cubic-anisotropic modifications to gravitational-wave dispersion at scales $E \sim M_P$, far below current observational sensitivity [21]. The D_4 framework predicts no observable Lorentz violation in gravitational-wave dispersion at $O((E/M_P)^2)$.

What remains open: all-loop renormalization, the cosmological-constant problem, the black-hole information paradox, nonperturbative or background-independent quantum gravity, and the geometric structure of strong-field configurations.

11 Conclusion

We have developed a self-contained discrete quantum-gravitational theory on the D_4 root lattice. The propagating graviton mode is identified with a coherent superposition of visons (m-type CSS excitations); this architectural choice is justified axiomatically (Theorem 1) by showing that the vison is the unique candidate among local discrete excitations satisfying the five graviton axioms. A single vison produces Regge deficit $\arccos(-1/3)$ at twelve surrounding edges (Proposition 2); coherent vison waves obey $\omega = c|\vec{k}|$ with masslessness (Proposition 4) and two transverse-traceless spin-2 polarizations (Proposition 5). Newton’s constant $G = a^2/(8 \ln 2)$ is fixed by Bekenstein-Hawking matching applied to the 2D sheet-plaquette area $L_0^2 = a^2/2$.

In the continuum limit, the Cheeger-Müller-Schrader theorem applied to the 4D simplicial structure of D_4 gives convergence of the discrete Regge action to the Einstein-Hilbert action (Theorem 7). The rank-four bond-tensor isotropy of D_4 (Theorem 8, proved self-contained in Section 8.3) ensures that the leading $O(a^2)$ corrections to the effective action are full $SO(4)$ -invariant curvature scalars (Theorem 9); the first cubic-anisotropic correction enters at $O(a^4)$ via the degree-six fundamental invariant of F_4 . The one-loop graviton self-energy, with momentum integration over the first D_4 Brillouin zone, is finite at each external momentum below the lattice cutoff (Proposition 10).

The picture across the gravitational sector of the SSM framework is internally consistent: classical Newton gravity from elasticity [5]; discrete graviton modes from vison coherent waves; continuum-limit recovery of nonlinear General Relativity from the Regge action on D_4 ; and lattice-regulated finite one-loop quantum gravity. What remains open—nonperturbative quantum gravity, the cosmological-constant problem, the black-hole information paradox—we have not solved, but the linearized and one-loop perturbative picture is now complete and matches standard GR by construction in the continuum limit.

Acknowledgments. This work builds on the FCC and D_4 lattice studies of [1–5] and the broader SSMTheory program.

Data availability. A Python script verifying all numerical claims of this paper (rank-2, 4, 6 tensor structure on D_4 , F_4 invariant theory, Brillouin zone bounds, Regge deficit angles, helicity decomposition) is available at https://github.com/raghu91302/ssmtheory/blob/main/verify_combined.py.

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